Synapse: A visible multi-agent pedagogical AI for Malaysian KSSM SPM science

by Team EduNova

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GitHub: github.com/Azimlearning/GenUI-Education

1.0 UNIVERSITI TEKNOLOGI PETRONAS INTRODUCTION

Established in 1997 and wholly owned by Malaysia's national oil and gas multinational corporation, PETRONAS, Universiti Teknologi PETRONAS (UTP) is a premier private university located on a sprawling 400-hectare campus in Bandar Seri Iskandar, Perak. Built with the vision of producing well-rounded global graduates with strong leadership and communication skills, the university currently serves a diverse student body of approximately 7,500 foundation, undergraduate, and postgraduate learners.

UTP offers a comprehensive array of industry-relevant programmes with a strong foundation in engineering, science, and technology. While globally renowned for its fully accredited engineering disciplines, including Petroleum, Chemical, Civil, Mechanical, and Electrical & Electronic Engineering, the university has significantly expanded its academic footprint. Today, it also provides robust degrees in Computer Science, Information Systems, Geosciences, and Business Management. This multidisciplinary approach allows students to bridge deep technical expertise with strategic, technopreneur thinking and analytical applications.

Operating as a research-intensive institution, UTP collaborates extensively with PETRONAS and other leading global industries to drive real-world applications and commercialisation. Its research framework is concentrated on six key niche areas: self-sustainable building, transport infrastructure, health analytics, hydrocarbon recovery, contaminant management, and autonomous systems. Through these specialised research institutes, the university actively champions the transition toward sustainable energy and contributes to the global goal of achieving Net Zero Carbon Emissions by 2050.

Over the years, UTP has cemented its position as a rising powerhouse in higher education, ranking in the 201–250 band in the Times Higher Education (THE) World University Rankings 2025 and 269th in the QS World University Rankings 2025. Its Petroleum Engineering programme consistently ranks among the top 20 globally. Beyond academic accolades, UTP boasts an exceptionally high graduate employability rate. This is driven by a curriculum tailored to industry demands and a mandatory, extended seven-month industrial internship programme that ensures students gain extensive, hands-on translational learning before entering the workforce.

2.0 INTRODUCTION

2.1 PROJECT BACKGROUND

Malaysia's science curriculum has evolved to prioritize Higher-Order Thinking Skills (KBAT), replacing rote memorisation with a demand for deep conceptual mastery. However, this approach assumes students have consistent access to hands-on laboratory practice, an assumption that fails in many rural and underfunded schools. Plagued by equipment shortages, large class sizes, and time constraints, countless SPM candidates are restricted to textbook diagrams or simple teacher demonstrations rather than actual experimentation. This creates a severe disconnect between what national exams demand and what students experience. Synapse bridges this gap by combining the urgent need for practical science education with the power of Generative AI. By turning the smartphones students already own into interactive virtual labs, Synapse democratises access to STEM, ensuring that every learner can develop the crucial conceptual foundations needed to succeed, regardless of their school's physical or financial resources.

2.2 IDENTIFIED PROBLEM

1. **Incompetency of traditional learning environments** rote-based teaching that doesn't build the conceptual mastery the KBAT curriculum now demands.
2. **Unequal access to crucial lab infrastructure** B40 and rural schools lack the apparatus and time to run the practical that build correct mental models.
3. **Persistent and hidden misconceptions** students can recite definitions correctly (e.g. osmosis, force and motion) while still holding well-documented, wrong underlying mental models that surface-level quizzes fail to catch.
4. **Inadequate EdTech and AI solutions** existing tools either generate generic content or act as opaque "black box" tutors, without transparent, verifiable pedagogical reasoning.

2.3 PROJECT OBJECTIVES

1. **Deliver targeted, pedagogically sound interventions** move beyond generic repetition by dynamically applying precise learning-science strategies (e.g. contrasting cases for misconceptions, scaffolded examples for knowledge gaps), ensuring students build the true conceptual mastery the KBAT curriculum requires.
2. **Provide universal access to faithful virtual experiments** deliver scientifically accurate, interactive virtual labs directly to the personal devices students already own, regardless of their school's funding or facilities.
3. **Diagnose root misconceptions with evidence-based precision** replace surface-level quizzes with a KSSM-aligned diagnostic tool checked against a well-researched database of known scientific misconceptions, pinpointing specific flawed mental models rather than guessing.
4. **Make AI reasoning transparent and auditable** eliminate the risk of AI hallucination and "black box" tutoring by streaming diagnostic and intervention choices live, building metacognitive awareness in students and giving teachers clear, verifiable oversight of every pedagogical decision.

3.0 PROPOSED SOLUTION

3.1 IDEA OVERVIEW

Synapse addresses this by giving every student a correct, virtual experiment paired with the right pedagogical intervention, on any device. Rather than a single model "generating a UI" for a topic which risks hallucinated or scientifically incorrect content Synapse uses a pipeline of specialised agents that:

1. **Diagnose** the student's specific conceptual state against a sourced KSSM misconception knowledge base (never inventing a misconception).
2. **Decide** the learning-science-correct intervention (e.g. contrasting cases, predict-observe-explain, worked-example fading, retrieval practice).
3. **Compose** a science-faithful interactive by selecting and configuring a pre-built component from a curated library the agents never generate UI markup from scratch.
4. **Track** the interaction over time, updating a persistent learner model and scheduling spaced repetition.

The differentiator is that this reasoning process is streamed and visible to the user in real time watching the Diagnostician, Pedagogy Strategist, and Component Composer "think" on screen rather than being hidden behind a single opaque generation step.

3.2 PROJECT SIGNIFICANT

- **Educational significance:** the misconception knowledge base is grounded in cited science-education literature (e.g. Odom & Barrow on osmosis, Taber on chemical bonding, Clement & Halloun-Hestenes on force, Driver on photosynthesis, Shipstone on circuits), and each strategy names its underlying learning-science technique.
- **Equity:** targets students in B40 and rural schools who lack reliable access to physical lab equipment.
- **Responsible AI:** no hallucinated science components are pre-built and faithfulness-pinned in code; the agents only select and configure, never freely generate.
- **Commercial path:** free for students, with a teacher dashboard (class-level mastery heatmap) as a B2B2C wedge toward school/district licensing.

4.0 TECHNICAL IMPLEMENTATION

4.1 REPOSITORY

GitHub: github.com/Azimlearning/GenUI-Education

4.2 ARCHITECTURE OVERVIEW

Synapse is a monorepo with two runtimes joined by a Server-Sent-Events (SSE) contract of typed component blocks:

Frontend Architecture

- **Core Stack:** Next.js (App Router), TypeScript, Tailwind CSS
- **Runtime/Environment:** bun
- **Primary Modules:**
  - Interactive Component Library
  - Server-Sent Events (SSE) Renderer

Backend Architecture

- **Core Stack:** Python 3.13, FastAPI, LangGraph, Pydantic v2
- **Primary Modules:**
  - 4-Agent System Architecture
  - Provider Router
  - Component Registry
  - Misconception Knowledge Base
  - Learner Store

The backend streams **typed component blocks** ({ pattern, props, meta }) over SSE; the frontend maps each pattern to a React component in its library and hydrates it with the streamed props. This schema is the load-bearing contract between the two runtimes.

4.3 THE AGENT PIPELINE

Agent	Role	Reads	Writes
Diagnostician	Identifies the student's specific misconception / knowledge gap / mastery level	Question + learner profile + misconception library	A diagnosis
Pedagogy Strategist	Selects the correct pedagogical intervention	Diagnosis	A strategy + target interaction pattern
Component Composer	Selects and configures a component from the library	Strategy + component registry	A typed component block (streamed via SSE)
Tutor Loop	Observes the interaction, updates the learner model, schedules spaced repetition	Interaction events	Learner-profile update

Data flows forward (question → diagnosis → strategy → composed component) and the tutor loop closes it backward (interaction → learner model → informs the next composition).

4.4 KEY TECHNICAL DECISION

- **Select, don't generate:** agents choose and parameterise pre-built, KSSM-faithful components rather than generating UI/markup from a model, eliminating hallucinated science by construction.
- **Interaction-pattern parameterisation:** components are built around ~12–18 reusable interaction patterns (e.g. one gradient/diffusion sandbox serves osmosis, dialysis, and gas exchange) rather than one bespoke component per topic this is what lets the library scale across the syllabus as configuration, not new code.
- **Multi-provider LLM router:** every LLM call goes through a config-driven provider router (Anthropic Claude as primary Haiku for the Diagnostician/Composer, Sonnet for the Strategist with OpenAI as fallback), with per-call cost/latency logging (GET /api/metrics). No agent calls a provider SDK directly.
- **Graceful degradation:** with no API key configured, agents fall back to scripted logic so the application always runs end-to-end; setting ANTHROPIC_API_KEY switches to live model reasoning.
- **Server-side keys only:** API keys are read only in the Python backend and are never exposed to the Next.js client.
- **Grounded diagnosis:** the Diagnostician classifies against an explicit, sourced misconception knowledge base (backend/app/knowledge/) rather than improvising a diagnosis freely.

4.5 TECH STACK

Backend: Python 3.13, FastAPI, LangGraph (agent orchestration), Pydantic v2 (typed agent state and component blocks), SSE via sse-starlette, a provider router over Anthropic and OpenAI, and a SQLite learner store (designed to be Firebase-ready behind the same interface).
Frontend: Next.js (App Router), TypeScript, Tailwind CSS, a pattern-based interactive component library (frontend/components/library/), an SSE client that renders streamed typed blocks, and bun as the package manager/runtime.

4.6 DEVELOPMENT STATUS

The pipeline is live end-to-end with LLM-driven agents (and a scripted fallback). All 12 target concepts render as real interactives, with three fully faithful flagship demos: osmosis, forces-motion, and bonding-electrons. The learner loop closes (mastery tracking + spaced repetition, informed by history), and a teacher dashboard is available at /teacher. The backend has 12 passing tests; the frontend typechecks and builds cleanly.

4.7 RUNNING THE PROJECT LOCALLY

```
# Backend
cd Synapse/backend
uv sync
cp .env.example .env          # optional: set ANTHROPIC_API_KEY for live model reasoning
uv run uvicorn app.main:app --reload --port 8000

# Frontend
cd Synapse/frontend
bun install
cp .env.local.example .env.local   # NEXT_PUBLIC_API_BASE=http://localhost:8000
bun dev                            # http://localhost:3000
```

5.0 COMMERCIAL VIABILITY

5.1 TARGET MARKET & STRATEGIC ENTRY

Synapse primarily targets the 380,000 to 400,000 Form 4 and Form 5 students undertaking SPM-level Physics, Chemistry, and Biology under the Malaysian KSSM curriculum annually. Our strategic entry point addresses the critical lack of laboratory access in rural and B40 communities. By transforming unfeasible physical practicals, such as titrations, ripple tanks, and ticker-tape experiments, into scientifically accurate, device-agnostic virtual simulations, we directly solve a major educational equity gap for under-resourced schools.

5.2 COMPETITIVE ADVANTAGE (THE MOAT)

Rather than relying on raw AI generation, which leaves startups vulnerable to well-resourced foundational models, Synapse's true competitive moat is its rigorously sourced misconception database paired with a faithful, KSSM-aligned component library. We have already mapped 12 concepts across three subjects using 14 reusable interaction patterns. Because these components are parameterized by interaction type rather than specific topics, scaling to new subjects requires simple configuration rather than entirely new code. Crucially, scientific accuracy is hardcoded into the platform, eliminating the risk of AI hallucinations that can destroy trust in educational technology.

5.3 GO-TO-MARKET & ADOPTION STRATEGY

- **Product-Led Growth (Bottom-Up):** A free tier for students designed to create immediate learning breakthroughs, driving organic, word-of-mouth adoption.
- **The B2B2C Wedge:** A dedicated teacher dashboard that distils the data of 40 students into actionable misconception heatmaps, incentivizing educators to champion the platform within their institutions.
- **Institutional Licensing (B2B):** Paid licenses for schools and fast-moving tuition centers, offering premium features such as cohort analytics, class management tools, and full Bahasa Malaysia localization.
- **Government & Ministry Partnerships:** Leveraging our alignment with national equity priorities and B40 lab accessibility to secure state or ministry-level pilot programs and scaled funding.

5.4 REVENUE MODEL

True to our equity-driven mission, the core student experience remains free; we monetize institutions, not disadvantaged learners. Revenue is generated through school and district-level licenses that unlock advanced teacher analytics, localized interfaces, and broader cohort reporting, with scalable volume discounts for larger districts.

5.5 LONG-TERM SUSTAINABILITY

Beyond its initial launch, Synapse is structured to serve as a robust academic and commercial vehicle. The foundational misconception database and evaluation frameworks are highly publishable research assets. Furthermore, the scalable pattern-library architecture is a valuable technical contribution that perfectly positions Synapse for a sustainable, open-source core supplemented by a premium hosted SaaS model well beyond the initial hackathon phase.

6.0 EDUCATIONAL SIGNIFICANCE

Synapse operates on a rigorously researched educational foundation rather than improvised logic. Its misconception database is directly anchored in established science-education literature, incorporating foundational studies such as Odom & Barrow on osmosis, Taber on chemical bonding, Clement and Halloun-Hestenes on force and motion, Driver on photosynthesis, and Shipstone on circuits. Furthermore, the system's Pedagogy Strategist avoids generic AI outputs; instead, it explicitly deploys proven, evidence-based instructional frameworks, including contrasting cases, predict-observe-explain, worked-example fading, and retrieval practice.

6.1 EVALUATION FRAMEWORK

- **Target Demographics:** 10 to 15 Form 4 and Form 5 students, with a deliberate focus on those from B40 and rural communities.
- **Study Design:** A within-subjects methodology featuring a pre-test, system intervention, immediate post-test, and a delayed post-test one week later. The pilot will focus on three flagship concepts: osmosis, force and motion, and chemical bonding.
- **Assessment Instrument:** A two-tier diagnostic test that requires students to provide both an answer and their underlying reasoning. This specifically isolates and identifies "right answer, wrong reason" scenarios—the exact cognitive gap Synapse is built to resolve.
- **Primary Metrics:** The core measure of success is the normalized gain in fully correct responses (accurate answer paired with accurate reasoning) between the pre- and post-tests. The one-week delayed check will specifically validate the effectiveness of the platform's spaced-repetition scheduling.
- **Control Group Comparison:** Where feasible, a static-explanation control group will be utilized to prove that the interactive, active-learning pedagogy (like predict-observe-explain) drives superior understanding compared to merely presenting factual information.

6.2 PRACTICAL CLASSROOM INTEGRATION

Synapse is engineered for seamless deployment in real-world educational ecosystems. It includes a comprehensive teacher dashboard for human oversight, allowing educators to track individual conceptual gaps and overall mastery. To ensure maximum accessibility, the platform features keyboard-navigable interactive components, planned localization into Bahasa Malaysia, and a lightweight, mobile-first architecture that runs effortlessly on the smartphones students already own, completely bypassing the need for specialized laboratory hardware.

7.0 REFERENCES

- refdocs/synapse-prd.md - full product requirements document
- refdocs/PROJECT-DESCRIPTION.md - hackAstone written submission
- refdocs/concept-catalog.md - the 12 demo concepts and their interaction patterns
- refdocs/EVALUATION-PLAN.md - pedagogical evaluation methodology
- refdocs/GO-TO-MARKET.md - commercial viability plan
- README.md - repository overview and run instructions